Preface: QuantConnect

QuantConnect provides a comprehensive environment for researching, backtesting, optimizing, and deploying live algorithmic trading strategies. Founded in 2012 it has a community of more than 300,000 registered quants, engineers, and data scientists.

QuantConnect supports 11 asset classes, including equities, futures, forex, options, and cryptocurrencies, in a cloud or on-premise environments. It provides a comprehensive data library, including price, fundamental, and alternative data.

It hosts a professional caliber, co-located, low-latency live-trading environment with direct connectivity to 20 brokerages and routing to more than 1,200 destinations (qnt.co/book-cloud-live-trading).

Transparency, freedom, and security are key focuses of the community ethos. Users own all their intellectual property (IP) and code, and can use the open-source version of QuantConnect, LEAN (https://www.lean.io), to run independently hosted strategies.

Maintaining equal access to the technology is an important part of QuantConnect's mission.

A robust framework helps create, evaluate, and launch trading algorithms much more efficiently. For video walk-throughs of QuantConnect, see qnt.co/book-videos.

Benefits of Using Frameworks

We wanted this book to focus on fully implemented algorithmic trading strategies, allowing our readers to wrestle with code and explore the markets for alpha, rather than learning about scaffolding or parsing data. By using a robust framework, all of the data loading and basic portfolio management are handled, saving you weeks to years of effort.

Efficiency

Frameworks offer hundreds of ready-made components and libraries, which save development time. These include basic alpha signals, portfolio construction systems, and open-source execution algorithms. Through building on top of a framework, you can leverage thousands of lines of code within a few statements.

Standardization

Frameworks provide demonstration template algorithms and a community of members familiar with the Application Program Interface (API).

Error Reduction

Leveraging frameworks will dramatically reduce the likelihood of coding errors and potential losses in production trading. No quant or engineer is perfect; for a critical system managing capital, you need something with a track record and reviewed by thousands of people. Frameworks generally have less errors as they are covered with unit and regression tests.

Focus on Strategy

The traditional process of building an algorithmic trading strategy devotes 95% of efforts to the same three activities: data parsing and loading, brokerage connectivity, and dashboards or user interfaces, none of which improve your investment performance. By harnessing a framework, traders can focus on their trading strategies instead of infrastructure.

QuantConnect Framework

QuantConnect serves an open-source algorithmic trading engine, LEAN, with an extensive library of financial data and cloud-computing resources. With minimal effort, you can start a quantitative trading technology stack and deploy strategies for production trading (see Figure 0.1).

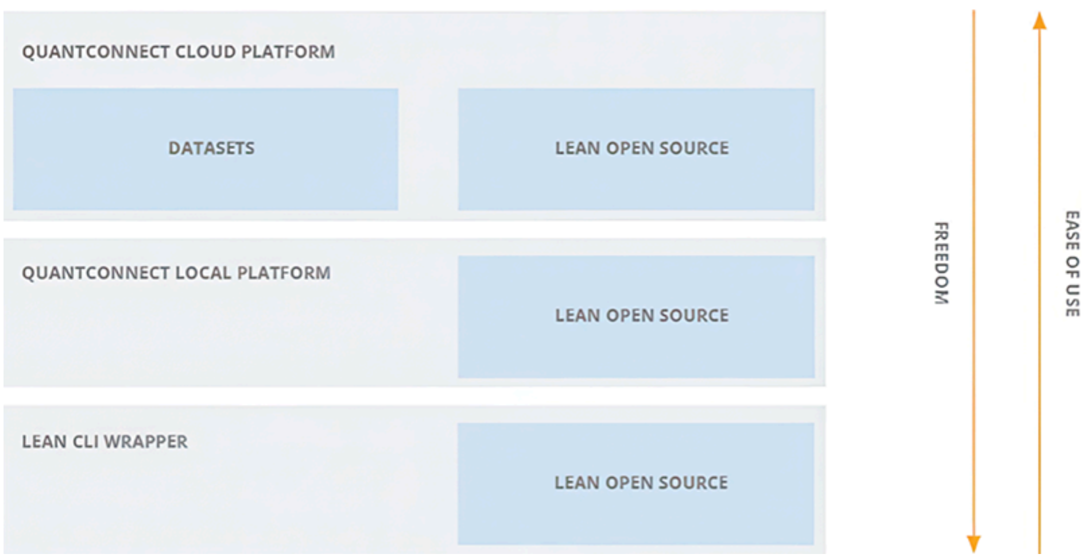

Figure 0.1 QuantConnect enables you to run LEAN via three platforms.

Its technology centers around three elements:

- Cloud Platform (qnt.co/book-cloud)—A cloud-hosted managed environment that lets your team focus entirely on alpha creation and trading. Data updates, cloud infrastructure, and team onboarding are all handled automatically.
- Local Platform (qnt.co/book-local)—An on-premises environment serving the same cloud user interface but giving you the freedom to add custom libraries, load proprietary datasets, and operate entirely on your organization's servers.

- LEAN CLI (qnt.co/book-cli)—A simple PIP package, CLI offers LEAN's entire power via an easy-to-use command line interface. Results are rendered as JSON objects for processing with your favorite charting program.

For all these platforms, the loading and parsing of data is handled automatically. There are integrations to historical and live-streaming data vendors and 60 alternative data vendors letting you focus on alpha research.

Creating an Account

To run most of the examples in this book, you will need an account on QuantConnect. Creating an account is free and takes less than 2 minutes. Register at quantconnect.com/signup.

Research

QuantConnect cloud-based research terminals attach to terabytes of financial, fundamental, and alternative data, preformatted and ready to use (Figure 0.2). Hosted alternative data are linked to the underlying securities, tagged with the FIGI, CUSIP, and ISIN to facilitate building strategies. Users can access popular machine learning and feature selection libraries to quantify factor importance, and install custom packages on request.

Figure 0.2 QuantConnect cloud-hosted Jupyter research environment.

Backtesting

With minimal-to-no code changes, clients can move from research to point-in-time, fee, slippage, and spread-adjusted backtesting on lightning-fast cloud cores (Figure 0.3). It is easy to perform multi-asset backtesting on portfolios comprised of thousands of securities with realistic margin-modeling.

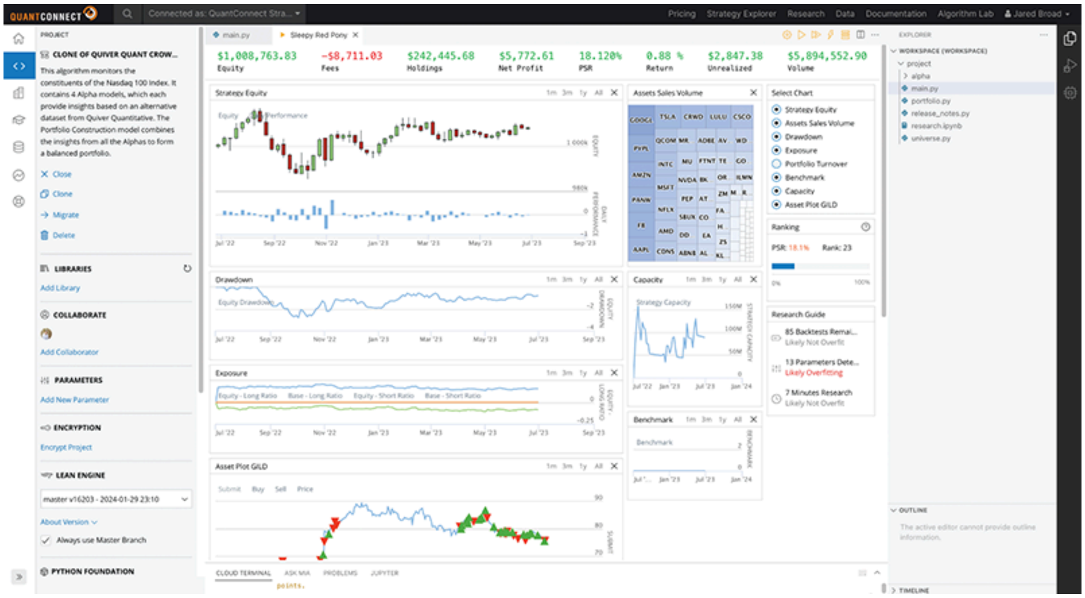

Figure 0.3 Accurate point-in-time backtesting, across 11 asset classes, down to tick resolution data.

You can import custom and alternative-data, linked to underlying securities, for realistically modeling live-trading portfolios and avoiding common pitfalls like look-ahead bias. More than 20,000 backtests are performed by the QuantConnect user-base daily.

Optimization

The parameter sensitivity testing allows you to run thousands of full backtests on scalable cloud compute, completing weeks of work in minutes. Visualize all the iterations of parameters on heatmaps to quickly understand your strategy's sensitivity to parameters for robust out-of-sample trading. Explore further by opening each result and seeing its individual trades and backtest logs to completely understand the source of your alpha, and how sensitive it is to parameter changes (Figure 0.4).

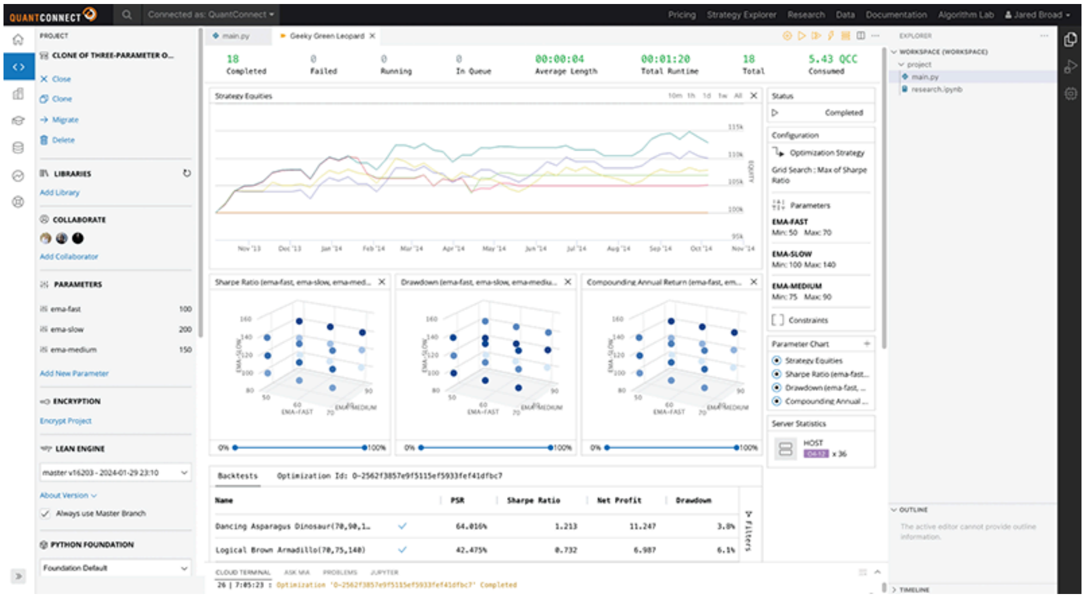

Figure 0.4 Scalable cloud parameter optimizer for rapid sensitivity testing.

Live Trading

QuantConnect has deployed more than 300,000 live strategies to a managed, co-located live-trading environment (Figure 0.5). The platform processes more than $45B in notional volume per month, serving hundreds of fund clients. Quants can execute trades directly through our 20 integrations, or route to 1,200 liquidity providers.

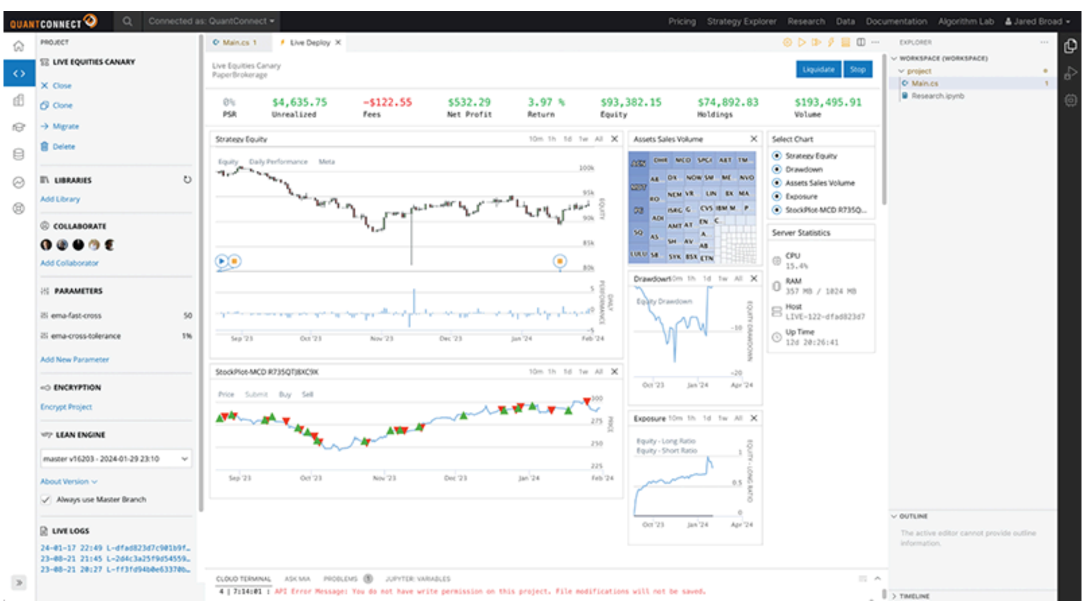

Figure 0.5 Instantly deploy strategies to live-trading with all data, user-interface, and order management handled automatically.

Powered by an Open-source Core

QuantConnect is powered by LEAN (qnt.com/book-lean), an open-source algorithmic trading engine designed to handle the entire life cycle of a trading strategy through research, backtesting, optimization, and live deployment.

LEAN has a permissive, commercially friendly license so you are free to download it and run it on your own servers. The LEAN engine is built in C# for the data parsing and loading, and bridges to python for the algorithm analysis.

Emphasizing its commitment to open-source, QuantConnect has built the LEAN CLI—a command line interface to run the research, back-testing, optimization, and live trading technology anywhere in the world.

Its high-level architecture diagram is illustrated in Figure 0.6.

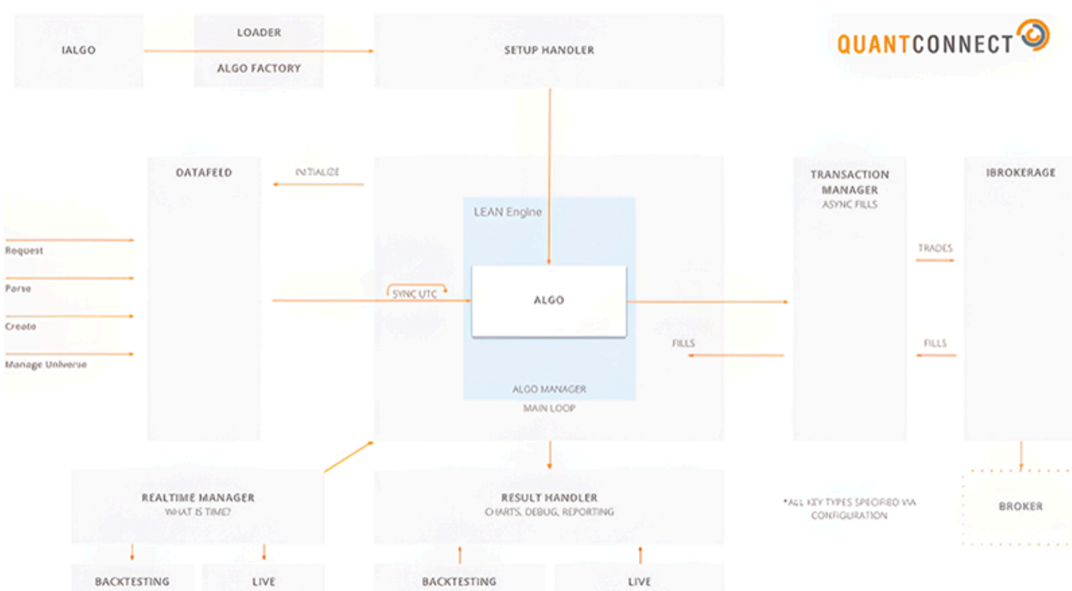

Figure 0.6 LEAN architecture.

The detailed description of the engine is outside the scope of this book, and we recommend that readers review the LEAN documentation (qnt.co/book-lean-docs).